Greedy and Math

Topic 9 of 10

2 questions | skeleton and writing space only

Each question: the prompt, the function or class signature, and ruled lines to write your own solution.

1. Wordle-Style 5-Letter Minimum-Guess Strategy

HARD | asked 2025-12-15 | algorithm-design, greedy

Phone screen coding: given a black-box oracle that tells you whether a specific letter is at a specific position in a hidden 5-letter word, design an algorithm to identify the word in as few oracle calls as possible. Hidden word: a 5-letter string from a..z. Oracle: query(pos: int, letter: char) -> bool returns whether position pos (0-indexed) holds letter. Return the hidden word. Primary metric: minimize the number of query calls.

```
def by_frequency(query):
```

2. Find All Bad Pairs Using runTest Oracle

HARD | asked 2025-11-27 | algorithm-design, math-reasoning

PhD intern coding round 2: given a black-box oracle runTest(S) that returns true iff a subset S contains no bad pair, identify all bad pairs across N test items with as few oracle calls as possible. Items t_1, ..., t_N are given. Oracle: runTest(S: Set[Item]) -> bool returns true iff S contains zero bad pairs; otherwise false. Output: every bad pair (t_i, t_j). Minimize total oracle calls. Examples: N=4, bad pairs = {(1,2)}.

```
def find_bad_pairs(items, run_test):
```